



Documentation:

Seed Of Hope

Client	The Seed of Hope Community Development NPO
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Module	INSY7315 — Work-Integrated Learning

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2.) Introduction

This document is submitted to the management of The Seed of Hope Community Development NPO (“Seed of Hope”) by our INSY7315 Work-Integrated Learning team — Chris, Andisa, Janaid and Reesaido — as the design and planning artefact for an e-commerce platform built on the organisation's behalf.

Seed of Hope runs a sewing programme producing handmade products including aprons, oven mitts, table runners, bags and cosmetic bags, alongside a separate children's after-school programme. The sewing programme's products are currently sold only through word of mouth and in-person events, which limits both reach and the income the programme can generate for its makers. Seed of Hope's Director, Heather Liebenberg, has asked our team to build an online shop so that these products can be listed, browsed and purchased by supporters over the web and from a mobile app, with every sale fairly paying the sewing programme for its work.

The purpose of this document is to record, in one place, the requirements we gathered from Heather, the design decisions our team has made in response to those requirements, and the plan by which we intend to deliver a working system by the end of the module. It is intended to give Seed of Hope's management visibility into what is being built and why, and to give our own team and assessors a single reference covering the full software development lifecycle: requirements, analysis, design, architecture, security, deployment, cost and change management.

3.) User Roles & Stories

User Roles

We identified three user roles for the platform, reflecting the baseline feature set the team agreed (browsing without an account, an account required at checkout, and a staff-managed catalogue).

- Guest — a visitor who has not logged in. Can browse the catalogue and hold items in a cart, but must register or log in to complete a purchase.
- Registered Customer — a supporter with an account. Can check out, view order history and track order status across whichever device they use (web or mobile), and manage their saved details.
- Staff / Admin — a Seed of Hope team member (e.g. Heather or a delegate) who manages the product catalogue, stock and orders.

User stories

- As a Guest, I want to browse the product catalogue without creating an account, so that I can decide what I'm interested in before committing to sign up.
- As a Guest, I want to add items to a cart, so that I can keep shopping before I decide to check out.
- As a Guest, I want to be prompted to create an account only at checkout, so that browsing stays frictionless.
- As a Registered Customer, I want to log in using biometric authentication on the mobile app, so that I can check out quickly and securely without typing a password each time.
- As a Registered Customer, I want to track the status of my order (received, packed, ready, collected), so that I know when to expect it.
- As a Registered Customer, I want to receive email notifications when my order status changes, so that I don't have to keep checking manually.
- As a Registered Customer, I want to reset my password if I forget it, so that I'm not locked out of my account.
- As a Staff/Admin user, I want to add, edit and deactivate product listings, so that the catalogue reflects what the sewing programme actually has available.
- As a Staff/Admin user, I want stock counts to update automatically after a sale, so that I don't accidentally sell an item that's no longer available.

5.) Work Breakdown Structure

Task	Description	Duration	Responsible	Resources Needed
1.1) Requirements & scope confirmation	Gather requirements from Heather; confirm scope, baseline features and NPO sign-off	2 weeks	Whole Team	Client interview, signed scope document
1.2) Analysis & design artefacts	UX journey map, domain & design class diagrams, sequence/state diagrams	2 weeks	Chris	Figma/UML tooling
1.3 Data & architecture design	ERD, technology stack justification, architecture & cloud diagrams, deployment plan	2 weeks	Reesaido, Janaid	Supabase docs, architecture diagramming tool
1.4 Security, DevOps & cost planning	Security/threat analysis, GitHub branching & CI pipeline design, running cost projections	1.5 weeks	Reesaido, Janaid	GitHub Actions, Supabase pricing pages
1.5 Documentation assembly & prototype	Merge all sections into one formatted document; build	1.5 weeks	Junaid, Andisa	Canva Template, Word
1.6 Task 1	Final proofread, Submission rubric self-assessment, submit as one PDF on ARC	2 days	Andisa, Junaid	ARC portal access
2.1 Backend & data setup	Provision Supabase project, schema, auth, storage buckets	2 weeks	Reesaido, Janaid	Supabase account
2.2 Web Application Build	Build web shop (catalogue, cart, checkout) and admin portal	3 weeks	Whole Team	React
2.3 Mobile Application Build	Build native Android app in Kotlin (catalogue, cart, checkout, biometric login)	3 weeks	Whole Team	Android Studio, Kotlin
2.4 Integration and Testing	Payment sandbox integration, automated tests, bug fixes	1.5 weeks	Andisa, Chris	Payment gateway sandbox
2.5 Task 2 Submission	Code freeze, final testing pass, submit repository/build	2 weeks	Chris, Junaid	GitHub repository
3.1 Final report & user guide	Consolidate final report, write end-user guide, prepare demo	Late October	Whole Team	Live Demo and Presentation

6.) Non-Functional Requirements

These are based on our requirements-gathering conversation with Heather and on the nature of the platform (a small NPO shop with modest but growing traffic, handling supporters' personal and payment information).

Category	Requirement
Performance	Product and catalogue pages should load in under ~3 seconds on an average mobile data connection. Supabase's managed Postgres and CDN-backed storage should comfortably meet this at the traffic levels expected in year one
Availability	The platform should target reasonable uptime for a small-NPO shop; Supabase's managed infrastructure removes the need for our team to run and patch our own database server
Privacy	Personal information collected at checkout (name, contact details, delivery/collection preference) must be handled in line with South Africa's Protection of Personal Information Act (POPIA) — collected only for the stated purpose, and not shared beyond what's needed to fulfil an order.
Compliance	Payment must be processed through a PCI-DSS compliant gateway (e.g. PayFast or Yoco); the platform itself should never store raw card details
Security	Passwords must be hashed, not stored in plaintext; the mobile app's biometric login must use the device's secure platform APIs (Android BiometricPrompt in Kotlin) rather than a custom implementation.
Usability	The admin/staff area must be usable by non-technical NPO staff with minimal training, since Seed of Hope has no dedicated IT staff.
Accessibility	Reasonable colour contrast and legible font sizes across both the website and the mobile app.

19.) Change Management

Why the organization needs to adopt this system

Seed of Hope currently has no digital sales channel; adoption isn't replacing an existing internal system so much as introducing a new capability. The main change-management task is therefore building staff confidence and habit around a tool they haven't used before, rather than migrating away from a legacy system.

Staff adoption strategy

- Identify one or two Seed of Hope staff members (likely Heather and a delegate) as the primary system users and points of contact during handover.
- Run a short, hands-on training session covering: logging into the admin portal, adding/editing a product listing, viewing and updating an order's status, and reading the basic sales dashboard.
- Provide a simple one-page quick-reference guide (screenshots plus short steps) that staff can keep on hand, in addition to the full user guide delivered with Task 3.
- Run the system in parallel with existing in-person/word-of-mouth sales for an initial period, so staff can build confidence before fully relying on it.

Customer / supporter adoption strategy

- Announce the shop through Seed of Hope's existing channels (social media, newsletter, word of mouth at events) once it's ready.
- Keep the checkout flow simple (guest browsing, account only required at checkout) to minimize friction for first-time buyers.
- Use email notifications to build trust in the ordering process (confirmation, status updates), since this is a new channel supporters haven't used with Seed of Hope before.

Handover plan (development → production)

Stage	What happens	Timing
Internal Testing	Team tests full purchase flow end-to-end using sandbox payments	During Task 2
Staff Training	Hands-on session with Heather / delegate on the admin portal	Early in Task 3
Soft Launch	Shop goes live with a small initial product list; team remains on call for issues	Task 3
Full Handover	Final user guide delivered; team support winds down to ad-hoc availability	End of module